# Springs, Solids and Fluids: Forces You Have to Measure

**SYNOPSIS**

Nine notes of this module have done the same thing: pick an object, draw the forces on it, apply ΣF = ma. This one is different in kind. It introduces three forces — a stretched spring, a squeezed solid, the air pushing back on a falling body — whose strengths **do not follow from Newton's laws at all**. Each carries a constant that nobody can derive and everybody has to measure. That is not a gap in the theory. It is how physics meets real materials.

**BEFORE YOU READ ON — PREDICT**

A skydiver steps out of an aircraft. From **N4** you know that near the Earth every falling object accelerates at 9.8 m/s², whatever its mass.

Does the skydiver still accelerate at 9.8 m/s² ten seconds later? Thirty?

Answer, with one sentence saying what would have to be true for your answer to be right, before you read on.

**PERFORMANCE EXPECTATIONS**

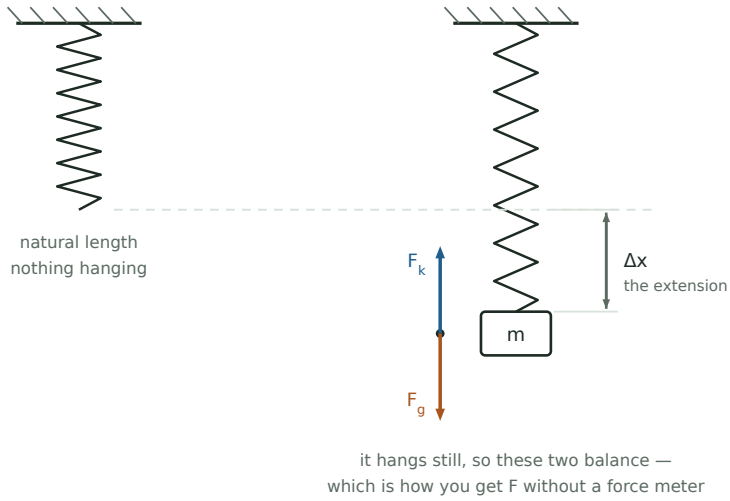
1. **Relate** the force in a stretched or compressed spring to its extension, and **distinguish** the extension from the spring's total length.
2. **Determine** k from the slope of a force-against-extension graph, state its unit, and **interpret** it as stiffness.
3. **Apply** the stress-strain relations for compression, stretch and shear, and **say** what Y and S describe.
4. **Explain** how terminal velocity arises from a force that grows with speed competing with one that does not.
5. **Compute** a terminal velocity from mass, area, shape and fluid density.
6. **Explain** why k, Y, S and $C_D$ have to be measured rather than derived, and **name** the constant from earlier in this module that is the same kind of quantity.

MOVEMENT 1 — A DIFFERENT KIND OF FORCE LAW

# Laws you derive, and laws you measure

MOVEMENT 2 — SPRINGS

## Hooke’s law



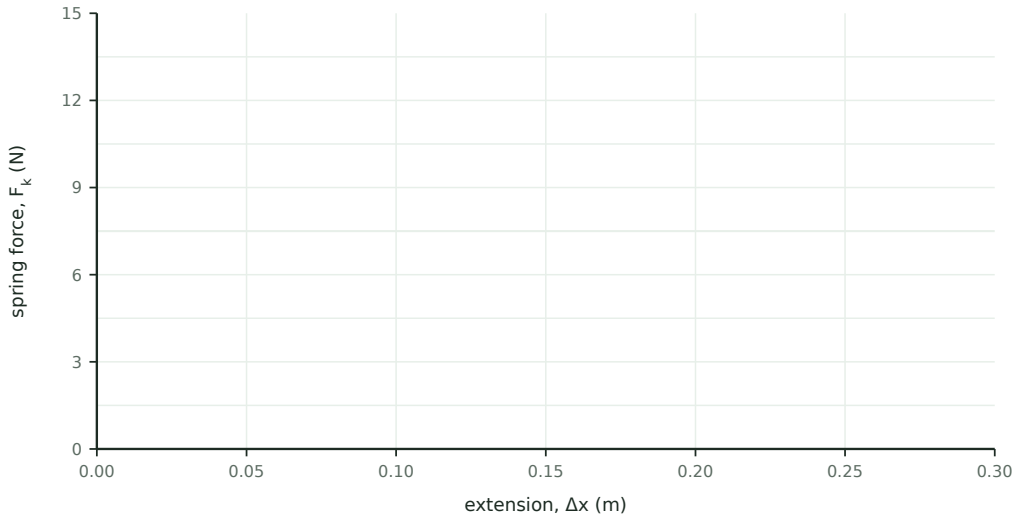

**Eq. 1** $F_k = k\Delta x$

double the extension and the spring pulls back twice as hard

## Measuring k

Hang known weights, record the extension each time, and plot one against the other. If Hooke’s law is right the points fall on a straight line through the origin, and the slope of that line is the spring constant.

| HANGING WEIGHT, F (N) | EXTENSION, ΔX (CM) | ΔX IN METRES |
|---|---|---|
| 2 | | |
| 4 | | |
| 12 | | |




Plot force against extension and draw the best straight line. Note the units on the horizontal axis — convert before you plot, not after.

1. Find the slope from your line, and give its unit.

2. You have two springs. One has k = 50 N/m and the other k = 500 N/m. Which is harder to stretch, and by what factor? Sketch both on the same axes.

3. A spring with a larger k gives a steeper line. Say in one sentence what “steeper” means physically here.

**WITH CALCULUS**

You found $k$ as the slope of a line you drew by eye. Written exactly, that slope is a derivative:

$$k = \mathrm{d}F_\mathrm{k} / \mathrm{d}(\Delta x)$$

Which is a stronger statement than it looks, because a derivative is defined at a *point* and your line was drawn across a whole range.

1. If the $F$–$\Delta x$ graph really is a straight line, what is true about $\mathrm{d}F_\mathrm{k}/\mathrm{d}(\Delta x)$ at every point on it? That statement is what “$k$ is a constant” actually means.

2. Sketch the graph for a spring pulled well past its elastic limit. Mark the stretch where $\mathrm{d}F_\mathrm{k}/\mathrm{d}(\Delta x)$ is constant, and say what is happening to that derivative beyond it.

3. So state the elastic limit as a condition on the **derivative** rather than on the spring.

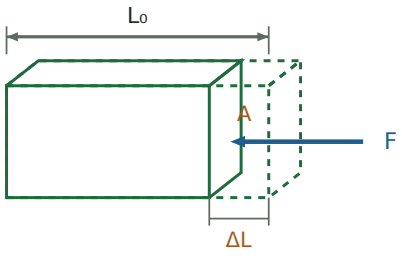
4. $Y$ and $S$ were defined the same way, as stress over strain. Write each as a derivative. What plays the part of $\Delta x$ in each one?

5. A constant obtained this way — the slope of a curve across the stretch where the curve is near enough straight — is called a **local linearization**. Using your sketch from task 2, say what the word *local* is doing in that phrase.

MOVEMENT 3 — SOLIDS

## Hooke’s law, spread over an area

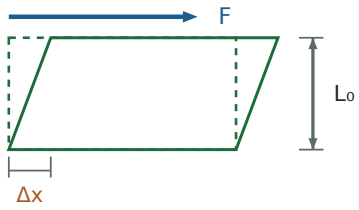
COMPRESSION AND STRETCH




A is the face the force pushes on

$F / A = Y (\Delta L / L_0)$

SHEAR




the top face slides; the block leans

$F / A = S (\Delta x / L_0)$

**IN BOTH CASES**

stress = modulus × strain

Y is the **Young’s modulus** and S the **shear modulus**. Both are properties of the material, not of the particular block — steel has the same Y whatever shape you cut it into, which is exactly why writing the law in terms of F/A and $\Delta L/L_0$ rather than F and ΔL was worth the trouble.

1. Compare $F = k\Delta x$ with $F/A = Y(\Delta L/L_0)$. Which symbol in the second plays the part that k plays in the first? What did the other two changes buy you?

2. Two bars of the same steel, same length, one twice the cross-sectional area. Under the same force, which shortens more, and by what factor?

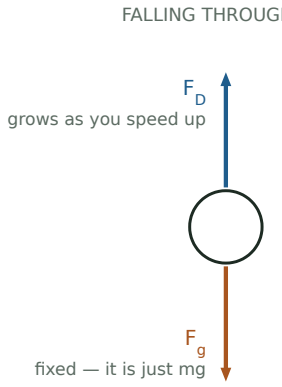
3. A material with a large Y — is it stiff or floppy? Give an example of each.

MOVEMENT 4 — FLUIDS

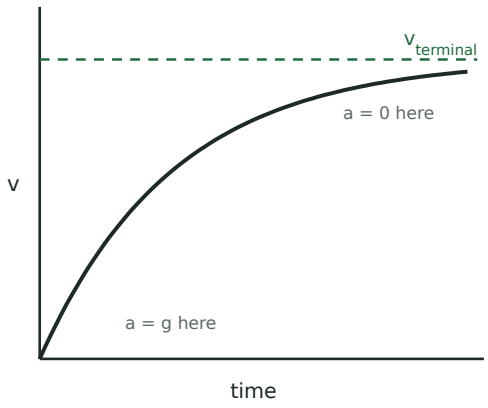
## Drag, and why falling stops speeding up

**Eq. 1** $F_D = \frac{1}{2} C_D \rho A v^2$

$C_D$ shape · ρ fluid density · A frontal area




the only force that changes is the drag




speed stops rising when the two forces match

4. Write ΣF = ma for the falling body, taking down as positive, with both forces in it.

5. Set the acceleration to zero and solve for v. This is the terminal velocity.

**Eq. 1** $v_{\text{term}} = \sqrt{\dfrac{2mg}{C_D \rho A}}$

the speed at which drag has grown to equal the weight

### TASK 6 — A SKYDIVER

1. Take m = 100 kg, frontal area A = 2 m², air density ρ = 1.3 kg/m³, and $C_D$ = 0.7 for a spread-eagled body. Find the terminal velocity in m/s.

2. Convert it to km/h.

3. Skydivers report roughly 200 km/h. Yours will not match. Is the disagreement small enough to call this model useful? Name **one** number in the calculation you would want to know better.

4. The same skydiver goes head-down, cutting A roughly in half. What happens to the terminal velocity — and by what factor, exactly?

### WITH CALCULUS

Everything so far has been about the *end* of the fall. The hook asked about the middle of it — whether the skydiver is still gaining speed after ten seconds, and after thirty. Answering that means writing the second law as an equation in d*v*/d*t* and solving it.

1. Take **down as positive**. Write Newton's second law for the falling skydiver with drag included, as an equation containing d*v*/d*t*.

2. Set d*v*/d*t* = 0 and solve for *v*. Compare with the $v_{\text{term}}$ you used in Task 6. What have you just shown about where that formula came from?

3. **Separate the variables and integrate**, taking $v = 0$ at $t = 0$, to show that $v(t) = v_{\text{term}} \tanh(gt / v_{\text{term}})$. Write your $v$-integral in terms of $v_{\text{term}}$ first — the standard form is easier to recognise that way.

4. Check your solution at the two ends: what is $v$ at $t = 0$, and what does $v$ approach as $t$ grows?

5. $v_{\text{term}}/g$ has units of time. Work it out from your own Task 6 answer. Roughly how many of those does the fall need to reach 99% of terminal velocity?

6. Sketch $v$ against $t$, marking $v_{\text{term}}$ and your timescale on the axes. Now answer the question in the hook from the graph rather than in words: still accelerating at ten seconds? At thirty?

**BACK TO YOUR PREDICTION**

You predicted whether the skydiver is still accelerating at 9.8 m/s² after ten seconds and after thirty. Was it right? Point to the moment on the speed-against-time graph where your answer changes, and name the force that changed it.

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